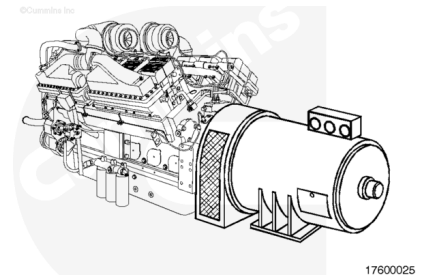


Trainings ▾

General Information

The PowerCommand Control 3200 and the generator-drive control system-equipped engines have multiple harnesses.

The engine harness is mounted on the engine and is routed from the inline connectors.

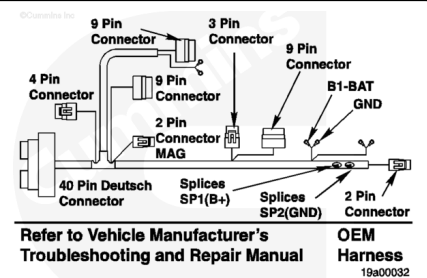


QST30

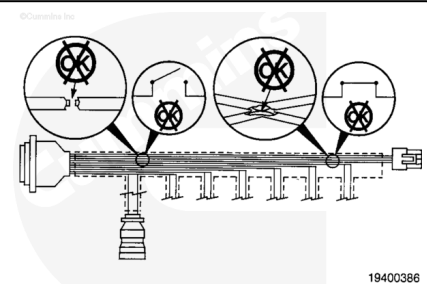
The QST30 generator-drive uses extension cables and an adapter cable to attach the engine harness to the engine control module (ECM).

The QST30 generator-drive upgrade engine also uses two separate wiring harnesses to control the engine and some of the equipment operations.

- Engine harness.
- Original equipment manufacturer (OEM) harness.



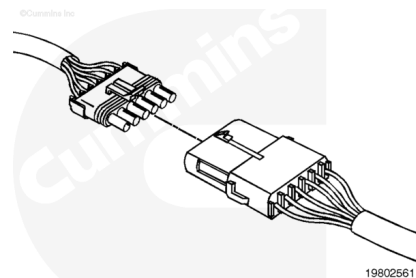
Replace a harness or extension/adapter cable if an open circuit or a short circuit is found under the protective covering of the harness body.



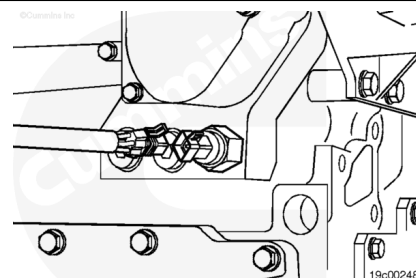
Remove

Qsx15

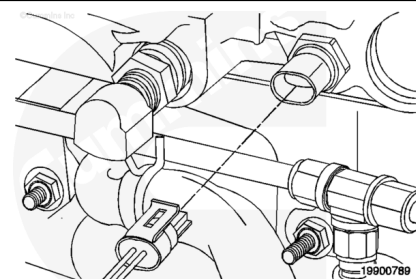
Disconnect the engine harness from the electrical system at the Inline E connector.



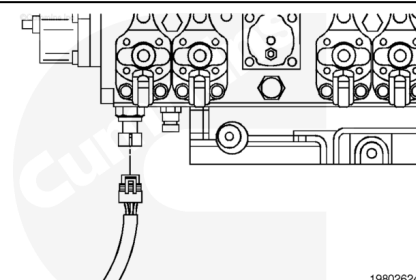
Disconnect the engine harness from the coolant temperature sensor.



Disconnect the engine harness from the intake manifold pressure/temperature sensor.

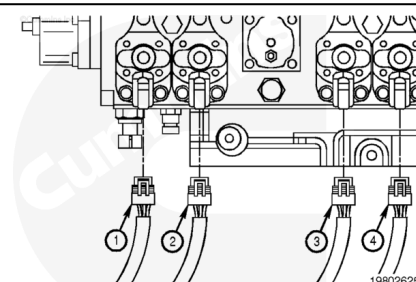


Disconnect the engine harness from the fuel pressure sensor.

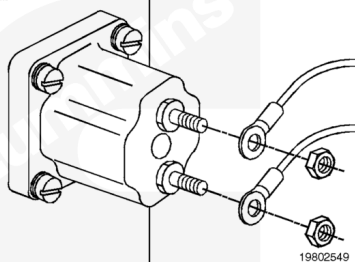


Disconnect the engine harness from the fuel system actuators:

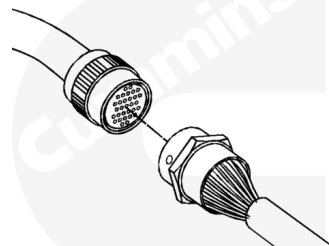
1. Fueling actuator 1.
2. Timing actuator 1.
3. Timing actuator 2.
4. Fueling actuator 2.



Disconnect the engine harness from the fuel shutoff valve.

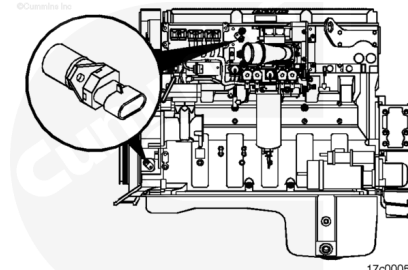


Disconnect the engine harness from the extension harness at Inline A, Inline B, and Inline C connectors.

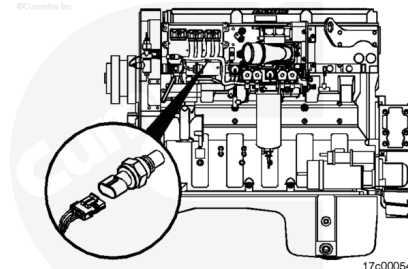


Disconnect the engine harness from the crankshaft engine speed sensor.

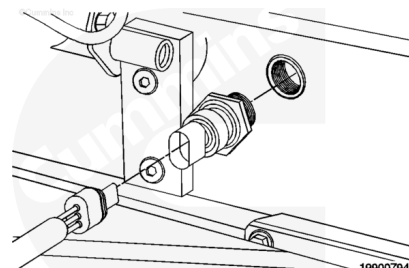
Disconnect the engine harness from the camshaft engine speed sensor.



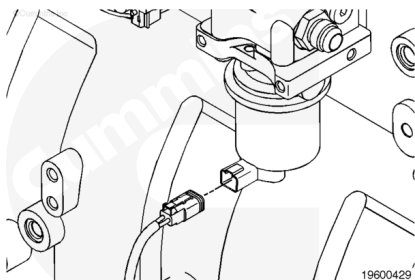
Disconnect the engine harness from the barometric pressure sensor.



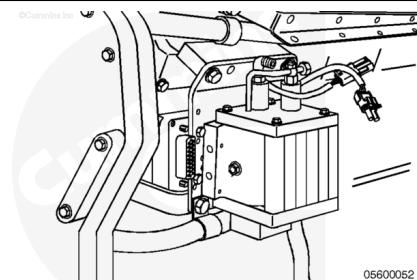
Disconnect the engine harness from the oil pressure/temperature sensor.



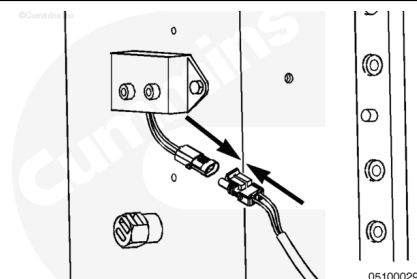
Disconnect the engine harness from the fuel supply pump connector, if equipped.



Disconnect the engine harness from the Centinel™ actuator connector, if equipped.



Disconnect the engine harness from the Centinel™ makeup tank oil level sensor inline connector, if equipped.

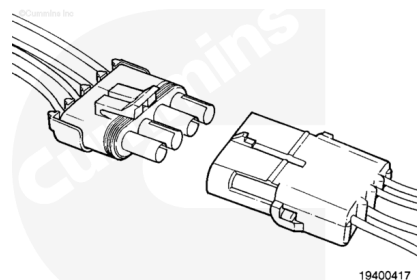


Disconnect the engine harness from the coolant level switch sensor at the 4-pin inline connector.

Remove any harness supports.

Make sure that all connections made by the engine harness are disconnected and untangled.

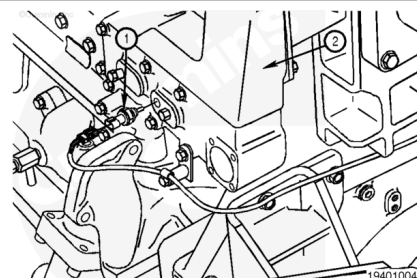
Slowly pull the engine harness away, and make sure that there is no binding or tangling.



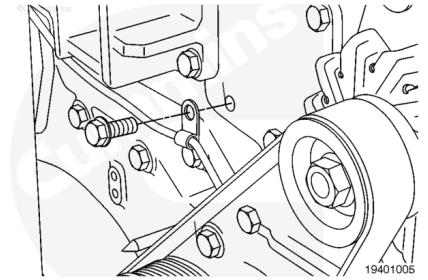
QSK23 Engines

Disconnect the engine harness from the coolant temperature sensor (1).

Remove the engine harness P-clip, spacer, and bracket mounted to thermostat housing (2).

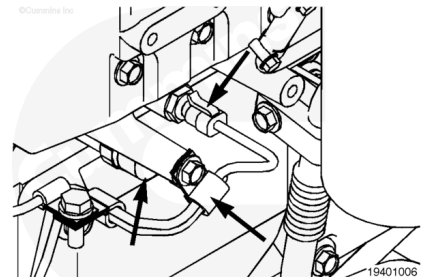


Remove the eight engine harness P-clips from the front of the engine, as shown.



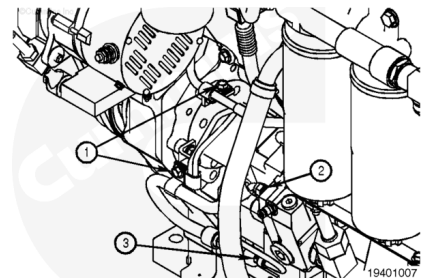
Disconnect the engine harness from the intake manifold temperature and pressure sensors, located on the right side of the fuel filters.

Remove the engine harness P-clip and spacer attached near the intake manifold pressure sensor.

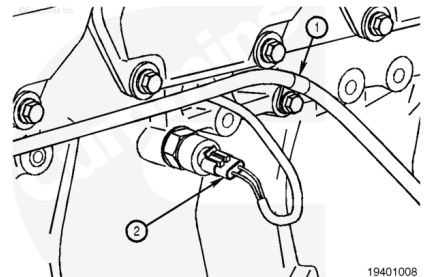


Remove the engine harness P-clips (1) attached to the fuel pump mounting flange.

Disconnect the engine harness from the fuel pump actuator (2) and fuel pump pressure sensor (3) connectors.

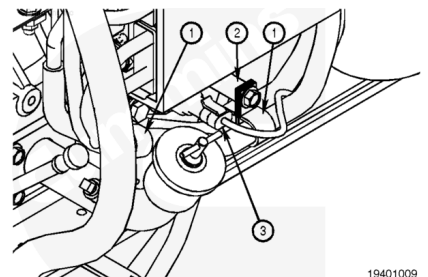


Remove the harness P-clip and the engine ground wire (1) and disconnect the oil pressure sensor (2).

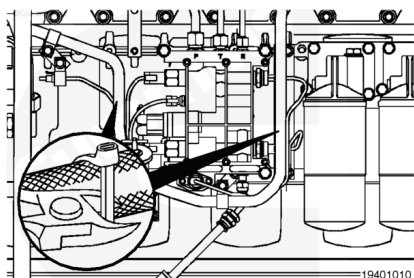


Remove the harness P-clips (1) and bracket (2) mounted to and just below the ECM near the ambient air pressure sensor (3).

Disconnect the ambient air pressure sensor (3).

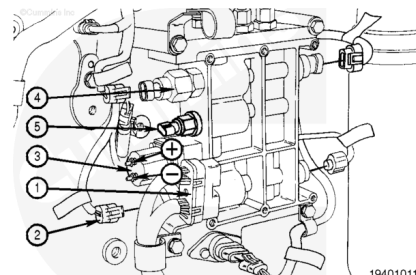


Cut the nine wire ties holding the engine harness to the support bracket.



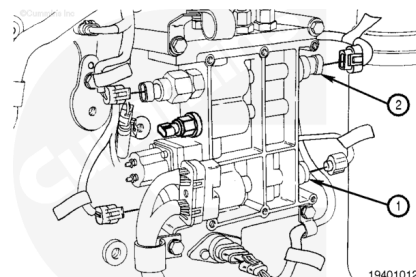
Disconnect the engine harness from the following:

1. ECM connector A.
2. Fuel rail actuator.
3. Fuel shutoff solenoid.
4. Fuel timing actuator.
5. Fuel temperature sensor.



Disconnect the engine harness from the following:

1. Rail pressure sensor.
2. Timing pressure sensor.

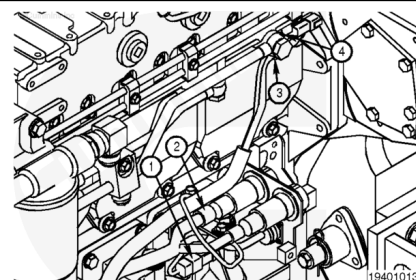


Note : The industrial wiring harness is shown, the power generation wiring harness routing will be slightly different and the harness will have some different connectors.

Disconnect the data link (1) connector connecting the engine harness to the OEM harness.

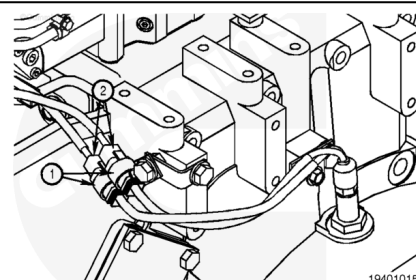
Remove the wiring P-clip (3) and intake heater connector bracket (4) located above the number 6 cam follower cover.

Remove the P-clip (2) near the lower right corner of the number 5 cam follower cover.



Remove the P-clips (1) to the rear of the intake manifold and the rear of the cylinder block.

Disconnect the engine speed sensor connectors (2) and remove the engine harness.



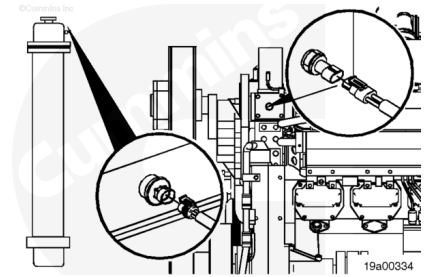
QST30

Disconnect the engine harness from the coolant pressure sensor, if equipped.

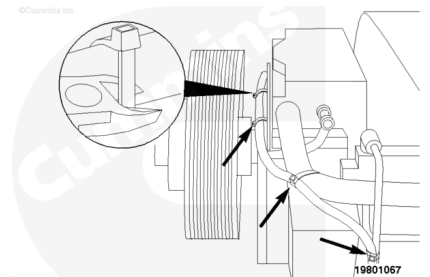
Remove the engine harness clamps.

Disconnect the engine harness from the coolant temperature sensor and coolant level sensors.

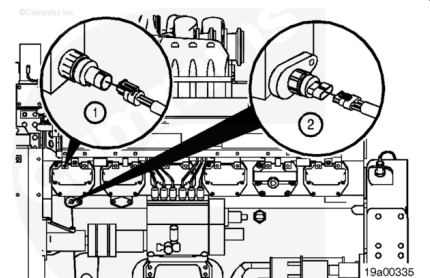
Disconnect the engine harness from the coolant level harness extension connector.



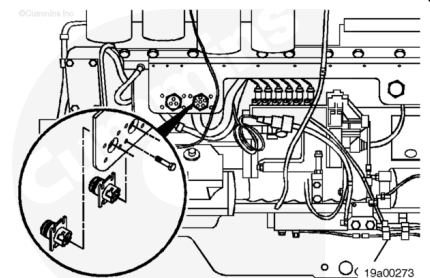
Cut the ties on the thermostat housing bracket, aftercooler tube, and fuel tube.



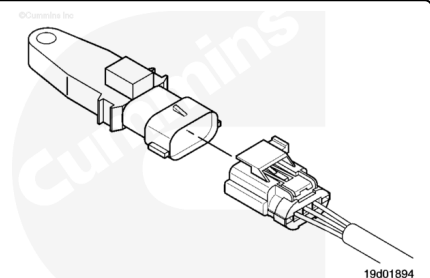
Disconnect the engine harness from the left bank intake manifold pressure sensor.



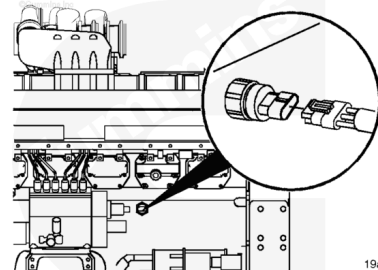
Disconnect the two 6-pin or one 9-pin data link connector(s) from the harness support bracket by removing the capscrews.



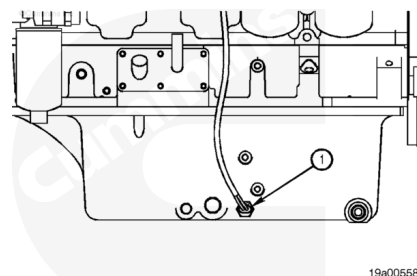
Disconnect the inline connector bracket which includes the barometric pressure sensor connector.



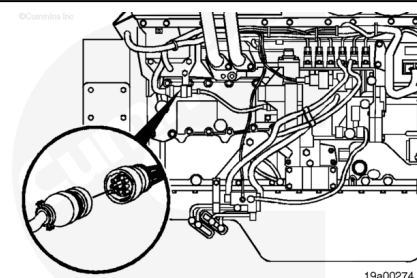
Disconnect the engine harness from the oil pressure sensor.



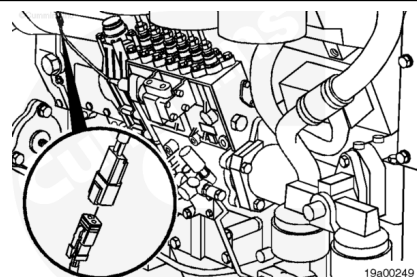
Disconnect the engine harness from the oil temperature sensor, if equipped.



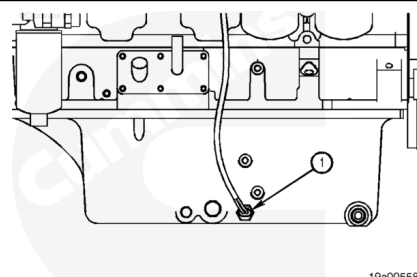
Disconnect the engine harness from the left bank fuel pump connector.



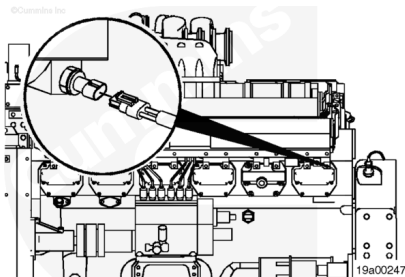
Disconnect the engine harness from the left bank fuel shutoff valve connectors.



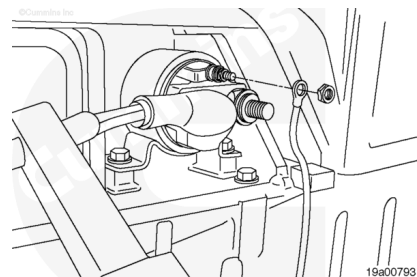
Disconnect the engine harness from the Centinel™ oil level extension harness, if equipped.



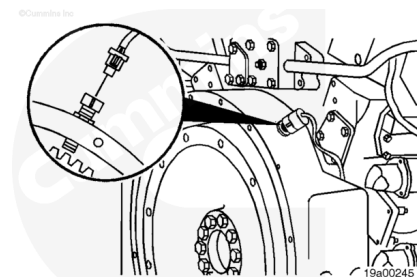
Disconnect the engine harness from the left bank intake manifold air temperature sensor.



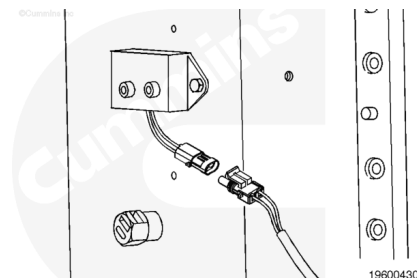
Disconnect the engine harness from the grid heater relay connector, if equipped.



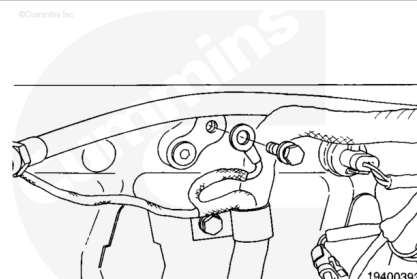
Disconnect the engine harness from the engine speed sensor.



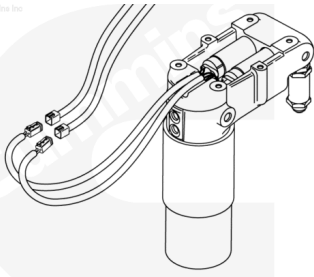
Disconnect the engine harness from the Centinel™ tank switch extension, if equipped.



Disconnect the engine harness engine block ground from the engine block.

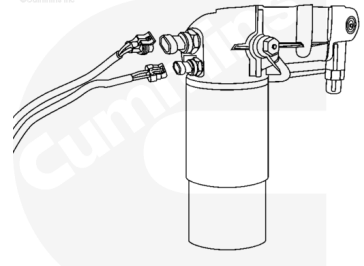


Disconnect the engine harness from the fuel supply pump connectors.



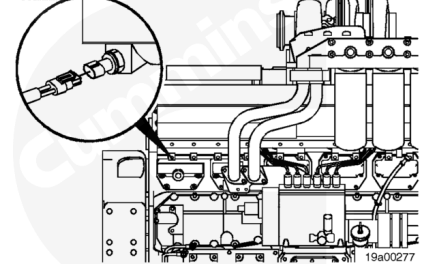
19a00792

Disconnect the engine harness from the fuel supply pump pressure and temperature sensor connectors.



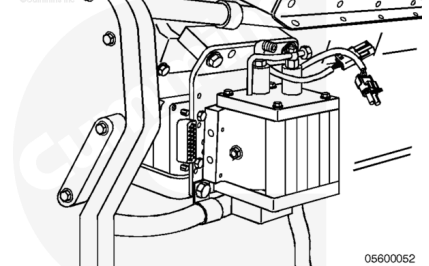
19a00791

Disconnect the engine harness from the right bank intake manifold air temperature sensor number 2 connector.



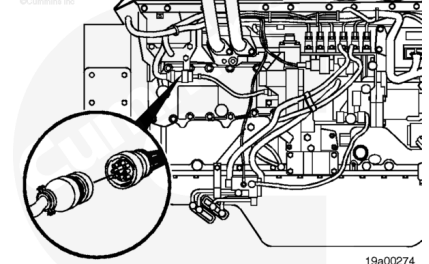
19a00277

Disconnect the engine harness from the Centinel™ actuator connector, if equipped.



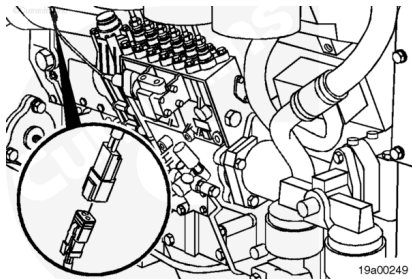
06600062

Disconnect the engine harness from the right bank fuel injection pump connector.



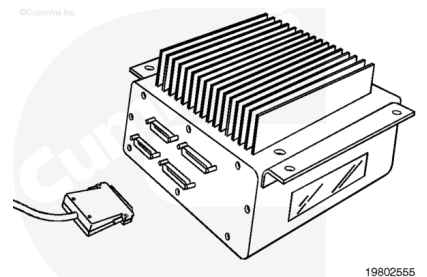
19a00274

Disconnect the engine harness from the right bank fuel shutoff valve connectors.



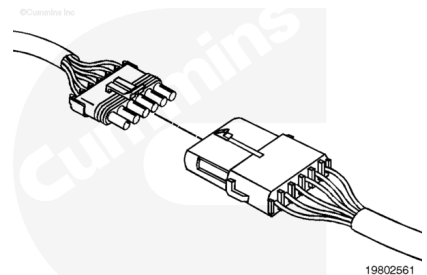
Remove all P-clips and harness retainers from the engine harness and remove the harness.

Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness Deutsch™ connector from the ECM.



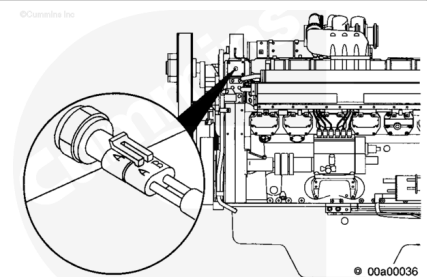
QST30 Upgrade

Disconnect the engine harness from the electrical system at the Inline E connector.

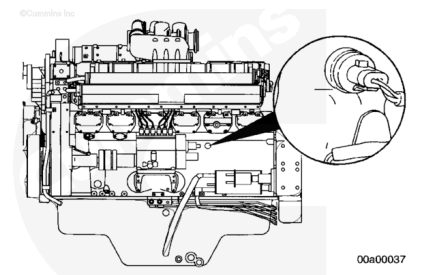


Remove the engine harness clamps.

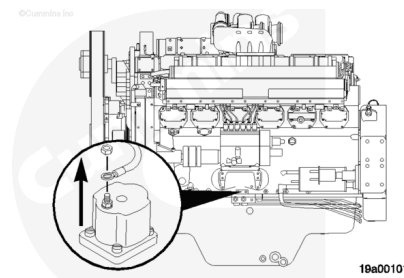
Disconnect the engine harness from the coolant temperature sensor.



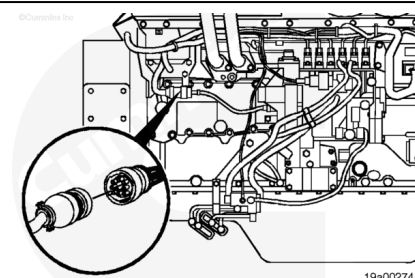
Disconnect the engine harness from the oil pressure sensor.



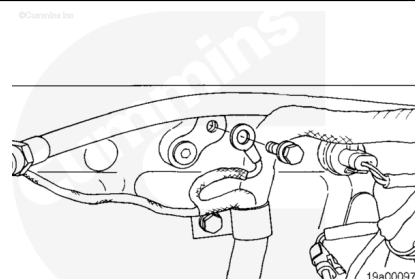
Disconnect the engine harness fuel shutoff control wires from the fuel shutoff solenoids (left bank and right bank).



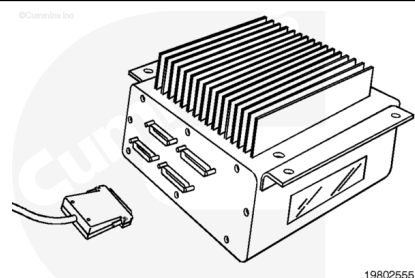
Disconnect the engine harness from the fuel pump connectors (left bank and right bank).



Disconnect the engine harness engine block ground from the engine block.

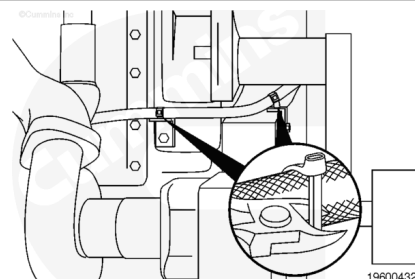


Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness Deutsch™ connector from any engine harness extension used or the engine harness adapter cable.



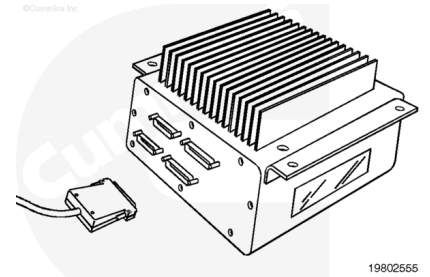
Cut the remaining ties holding the engine harness to the engine.

Remove the engine harness.



Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness extension cable Deutsch™ connectors.

Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness adapter cable Deutsch™ connectors from the ECM.

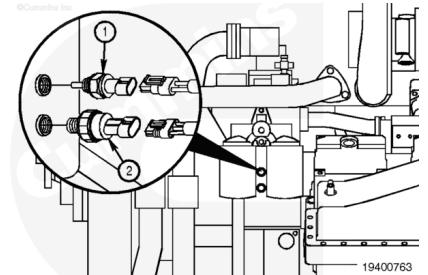


QSK45, QSK60, and QSK78

Note : The following graphics represent the QSK60 engines. However, QSK45 and QSK78 engines are similar.

The coolant pressure sensor may be located in one of three locations:

- Between the coolant filters, as illustrated.
- In the engine block, behind the fuel pump.
- In a spacer block, between the coolant filters and thermostat housing.

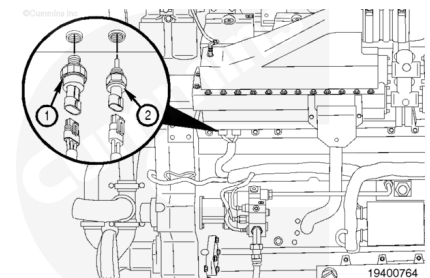


Disconnect the engine harness from the coolant temperature sensor (1).

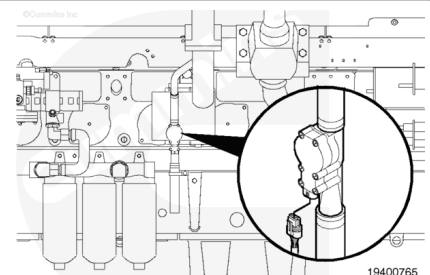
Disconnect the engine harness from the coolant pressure sensor (2).

Disconnect the engine harness from the intake manifold air pressure sensor (1).

Disconnect the engine harness from the intake manifold air temperature sensor (2).

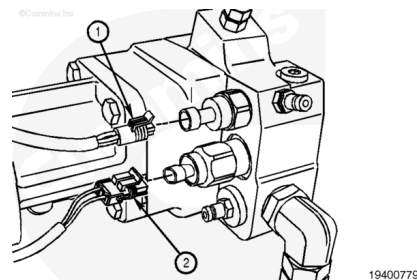


Disconnect the engine harness from the blowby pressure sensor.



Disconnect the engine harness from the fuel pump pressure sensor (1).

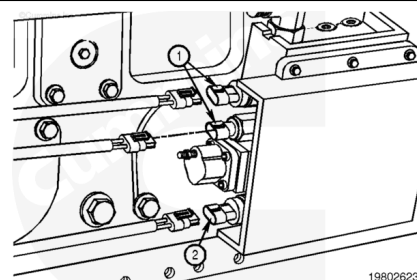
Disconnect the engine harness from the fuel pump actuator (2).



19400779

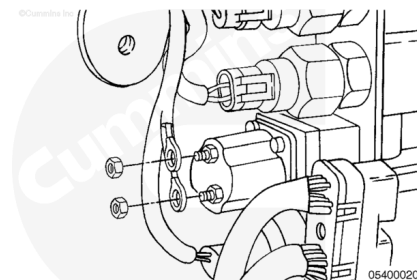
Disconnect the engine harness from the two timing rail actuators (1).

Disconnect the engine harness from the fueling rail actuator (2).



19802623

Disconnect the engine harness from the fuel shutoff valve.



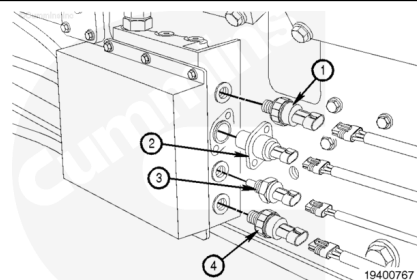
05400020

Disconnect the engine harness from the timing rail pressure sensor (1).

Disconnect the engine harness from the fuel temperature sensor (3).

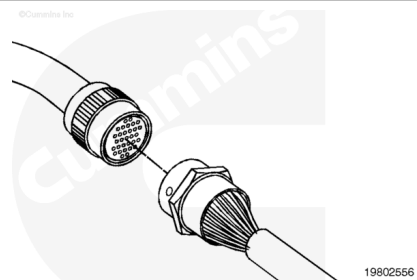
Disconnect the engine harness from the injector metering rail pressure sensor (4).

Disconnect the engine harness from the barometric pressure sensor (2). This can **not** be located on the electronic control valve assembly. Locate the sensor on the engine harness main trunk support bracket.



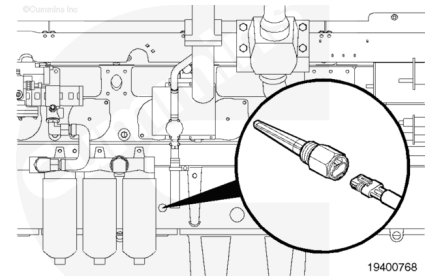
19400767

Disconnect the engine harness from the extension harness at Inline A, Inline B, and Inline C connectors.



19802556

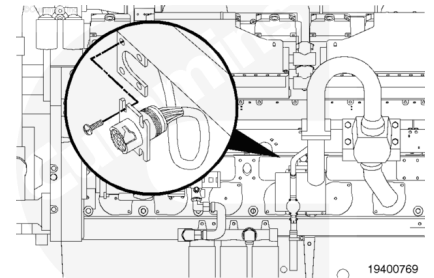
Disconnect the engine harness from the oil level sensor, if equipped.



Note : This step can not be completed on the QSK78 engine. The engine harness and engine support bracket are one piece.

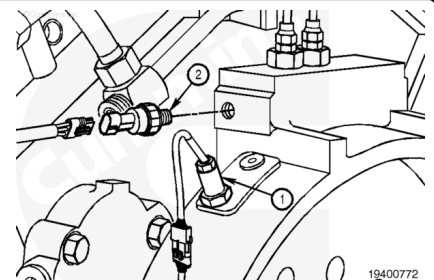
To disconnect the 9-pin data link connector from the engine harness support bracket:

1. Remove four capscrews.
2. Slide the 9-pin data link connector out of the mounting slot in the engine harness support bracket.



Disconnect the engine harness from the engine speed sensor (1).

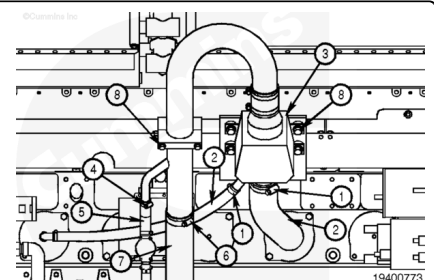
Disconnect the engine harness from the oil pressure sensor (2).



Note : There are a number of different versions of the breather box QSK45, QSK60, and QSK78 engines. Even though the appearance can be different, the same steps can be used to remove all versions of the breather box.

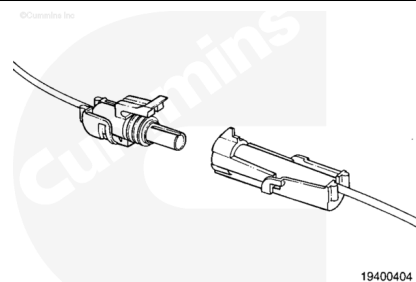
To remove the breather box assembly:

1. Loosen the three hose clamps (1) and disconnect the three hoses (2) on the bottom of the breather box (3).
2. Loosen the hose clamp (4) and disconnect the blowby pressure sensor hose (5).
3. Loosen the hose clamp (6) and disconnect the lower breather hose (7).
4. Remove the six attaching capscrews (8) and remove the breather box assembly.

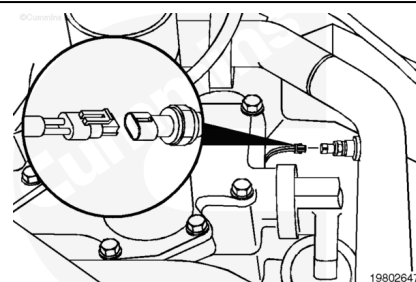


Remove the six attaching capscrews (8), and remove the breather box assembly.

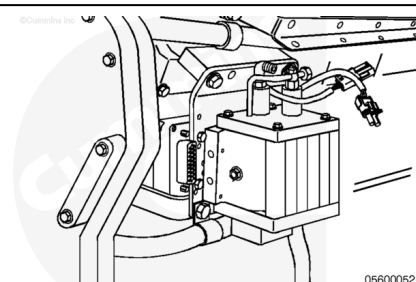
Disconnect the engine harness from the flash status at the Inline H connector.



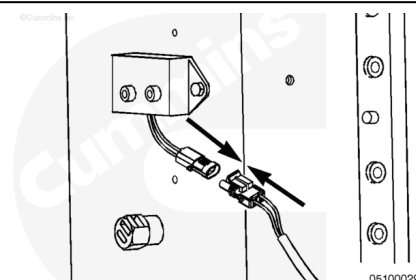
Disconnect the engine harness from the aftercooler water temperature sensor.



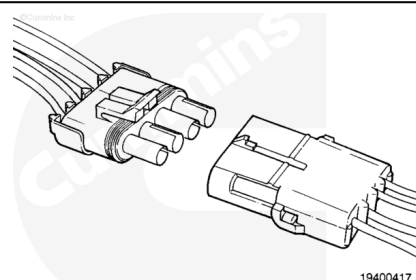
Disconnect the engine harness from the Centinel™ actuator connector, if equipped.



Disconnect the engine harness from the Centinel™ makeup tank oil level sensor inline connector.

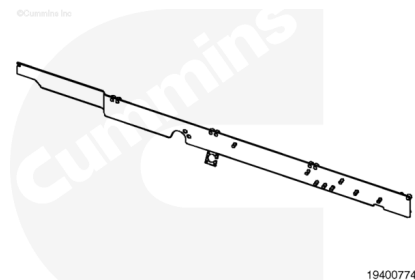


Disconnect the engine harness from the coolant level switch sensor at the 4-pin inline connector.



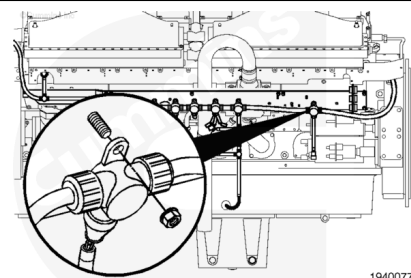
Disconnect the engine harness and engine harness support bracket from the engine.

Make sure that all connections on the engine harness are disconnected and **not** tangled.



Note : This step can **not** be completed on the QSK78 engine. The engine harness and engine support bracket are one piece.

Slowly pull the engine harness away from the engine block making sure the wires do **not** bind or tangle.



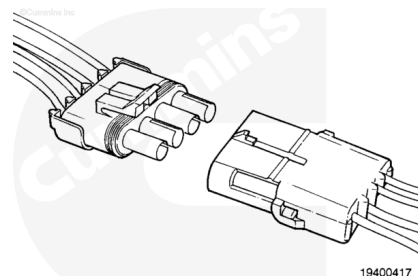
Install

QSX15

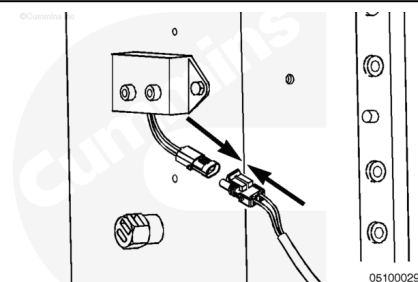
Place the engine harness on the engine.

Fasten the engine harness to the engine at the harness support.

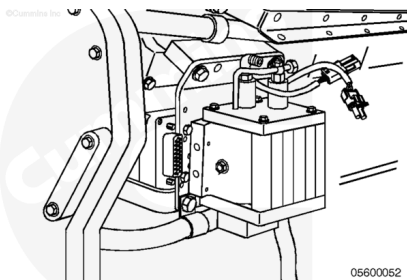
Connect the engine harness to the coolant level switch sensor at the 4-pin inline connector.



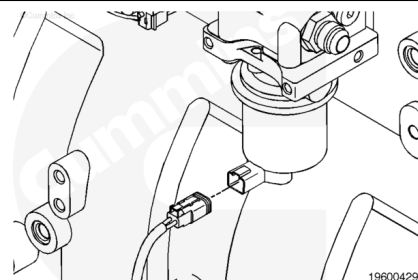
Connect the engine harness to the Centinel™ makeup tank oil level sensor inline connector, if equipped.



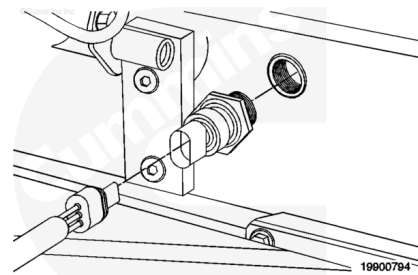
Connect the engine harness to the Centinel™ actuator connector, if equipped.



Connect the engine harness to the fuel supply pump connector, if equipped.



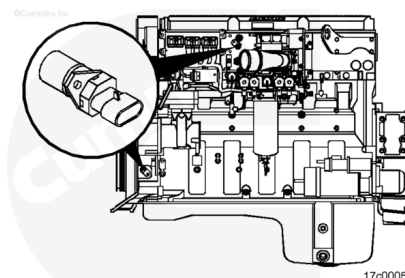
Connect the engine harness to the oil pressure/temperature sensor.



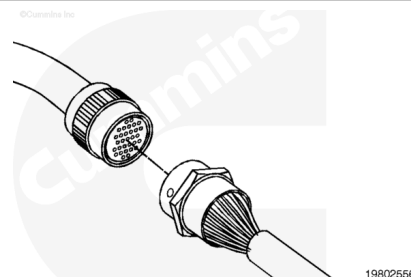
Connect the ambient air pressure sensor to the engine harness.

Connect the engine harness to the camshaft engine speed sensor.

Connect the engine harness to the crankshaft engine speed sensor.

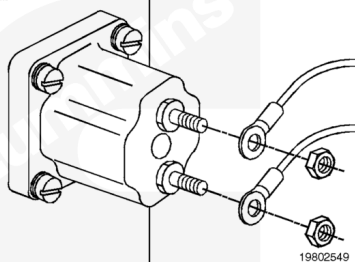


Connect the engine harness to the extension harness at Inline A, Inline B, and Inline C connectors.



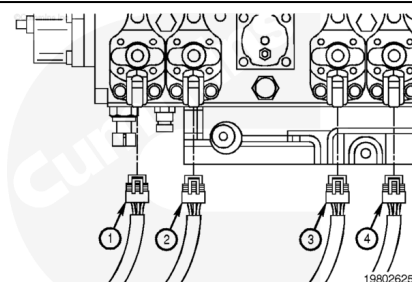
Connect the engine harness to the fuel shutoff valve.

Torque Value: 3 n·m [27 in-lb]

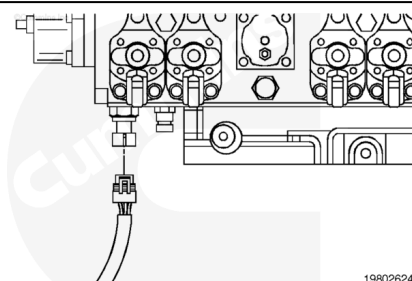


Connect the engine harness to the fuel system actuators:

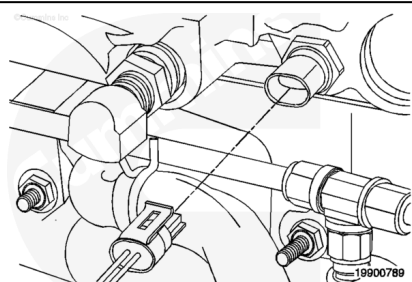
1. Fueling actuator 1.
2. Timing actuator 1.
3. Timing actuator 2.
4. Fueling actuator 2.



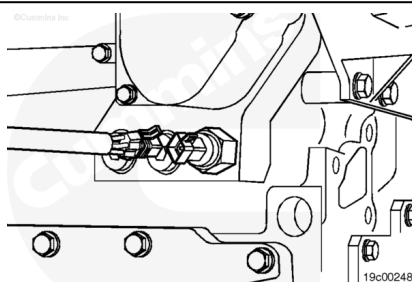
Connect the engine harness to the fuel pressure sensor.



Connect the engine harness to the intake manifold pressure/temperature sensor.



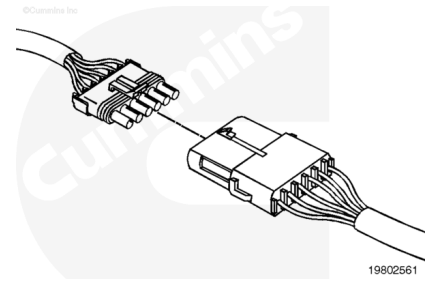
Connect the engine harness to the coolant temperature sensor.



Connect the engine harness to the Inline E connector.

Make sure that all engine harness connections are secure.

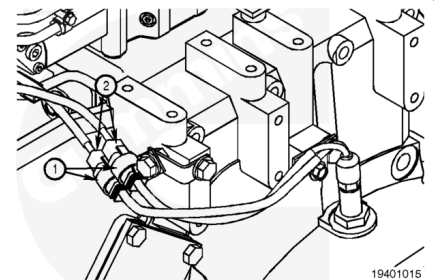
Connect the electronic service tool and check for fault codes.



QSK23 Engines

Install the engine harness and connect the engine speed sensor connectors (2).

Install the P-clips (1) to the rear of the intake manifold and the rear of the cylinder block.

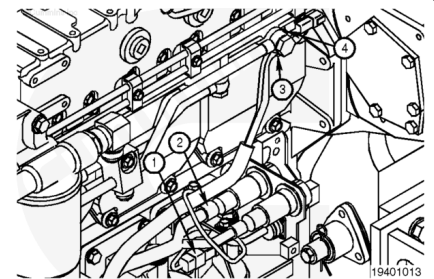


Note : The industrial wiring harness is shown. The power generation wiring harness routing will be slightly different and the harness will have some different connectors.

Install the P-clip (2) near the lower right corner of the number 5 cam follower cover.

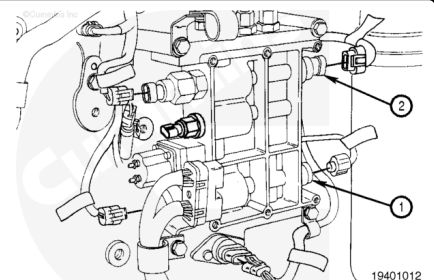
Install the wiring P-clip (3) and intake heater connector bracket (4) located above the number 6 cam follower cover.

Connect the data link (1) connector connecting the engine harness to the OEM harness.



Connect the following to the engine harness:

1. Rail pressure sensor.
2. Timing pressure sensor.

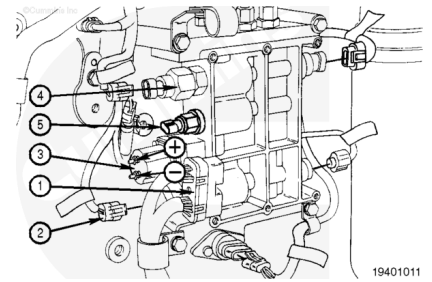


Connect the following to the engine harness:

1. ECM connector A.
2. Fuel rail actuator.
3. Fuel shutoff solenoid.
4. Fuel timing actuator.

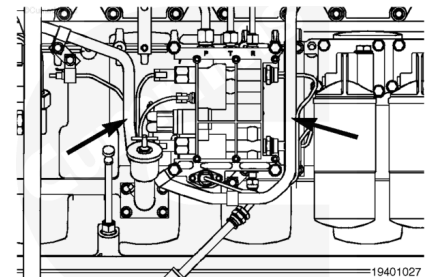
5. Fuel temperature sensor.

Make sure the positive (+) and negative (-) terminals on the fuel shutoff solenoid are correct.



19401011

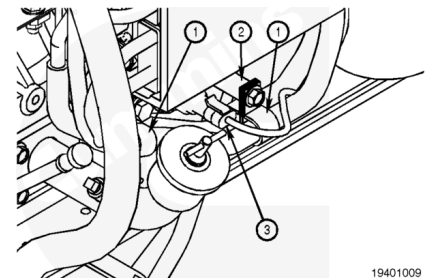
Install the nine wire ties holding the engine harness to the support bracket.



19401027

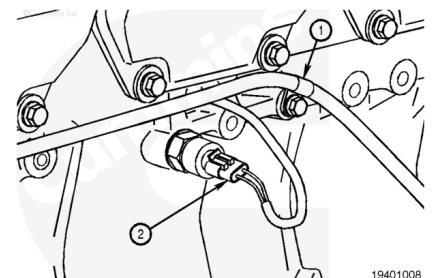
Connect the engine harness to the ambient pressure sensor (3).

Install the harness P-clips (1) and bracket (2) mounted to and just below the ECM near the ambient pressure sensor (3).



19401009

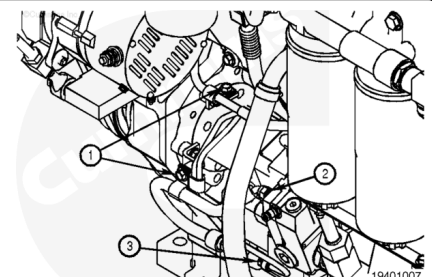
Install the harness P-clip and engine ground wire (1) and connect the oil pressure sensor (2).



19401008

Connect the harness to the fuel pump actuator (2) and fuel pump pressure sensor (3).

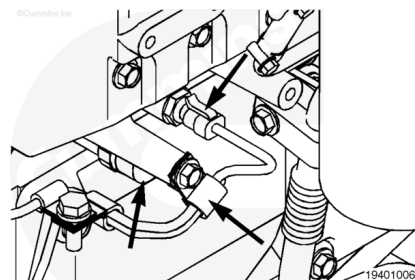
Install the engine harness P-clips (1) attached to fuel pump mounting flange.



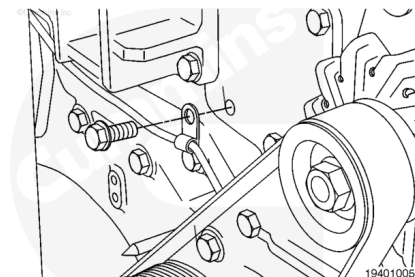
19401007

Install the engine harness P-clip and spacer attached near the intake manifold pressure sensor.

Connect the engine harness to the intake manifold temperature and pressure sensors, located on the intake manifold on the right hand side of the fuel filters.

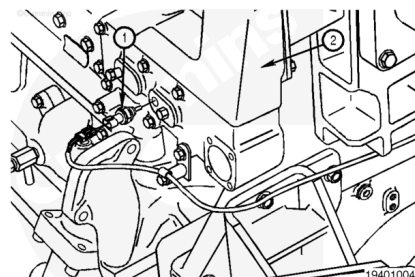


Install the eight engine harness P-clips on the front of the engine, as shown.



Install the engine harness P-clip, spacer, and bracket mounted to the thermostat housing (2).

Connect the engine harness to the coolant temperature sensor (1).

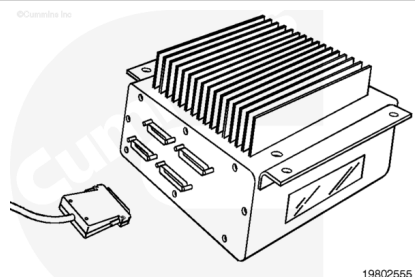


QST30

Install all P-clips and harness retainers to the engine harness

Use a 4 mm [5/32 in] hex head wrench to connect the engine harness Deutsch connector to the ECM.

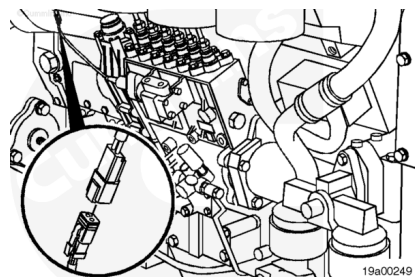
Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.



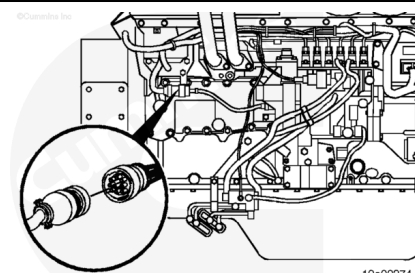
Torque Value: 2.8 n·m [25 in-lb]

Note : Do not over-torque as connector damage can occur.

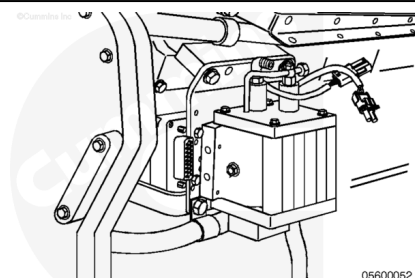
Connect the engine harness to the right bank fuel shutoff valve connectors.



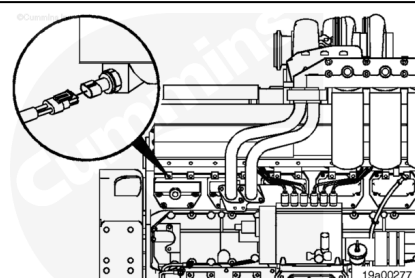
Connect the engine harness to the right bank fuel injection pump connector.



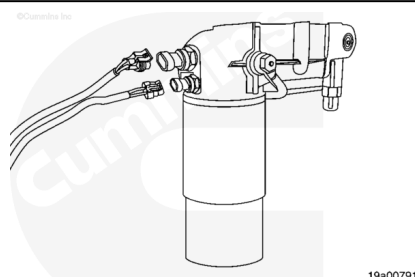
Connect the engine harness to the Centinel™ actuator connector, if equipped.



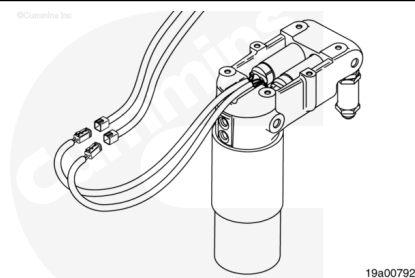
Connect the engine harness to the number 2 right bank intake manifold air temperature sensor connector.



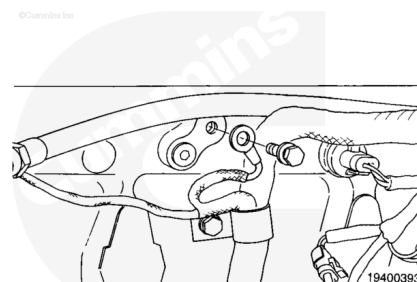
Connect the engine harness to the fuel supply pump pressure and temperature sensors.



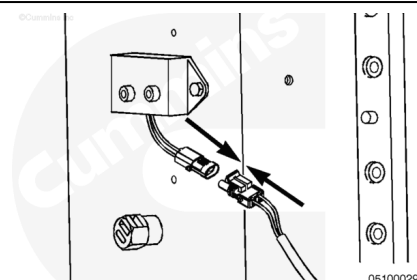
Connect the engine harness to the fuel supply pump connectors.



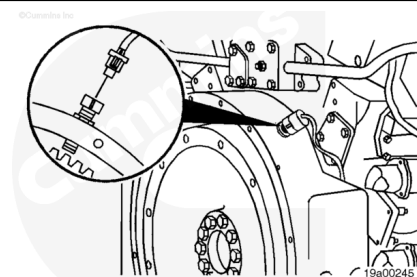
Connect the engine harness engine block ground to the engine block.



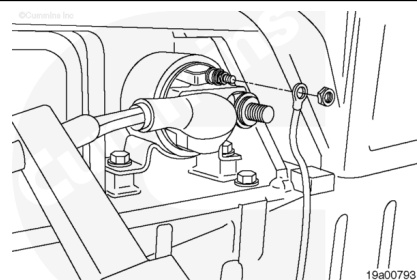
Connect the engine harness to the Centinel™ tank switch extension, if equipped.



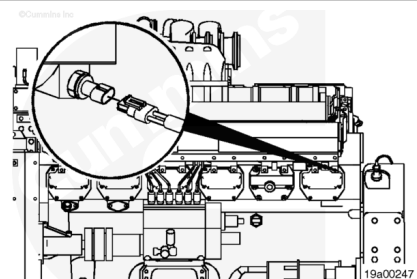
Connect the engine harness to the engine speed sensor.



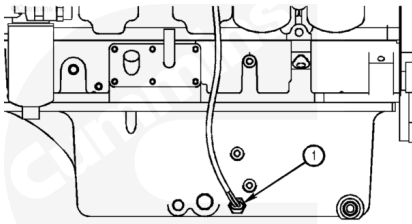
Connect the engine harness to the grid heater relay connector, if equipped.



Connect the engine harness to the left bank intake manifold air temperature sensor.

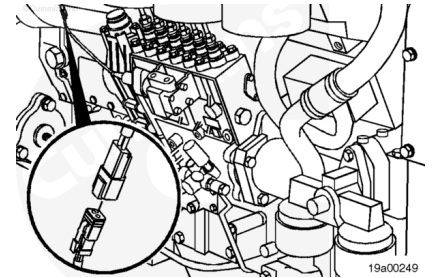


Connect the engine harness to the Centinel™ oil level extension harness, if equipped.



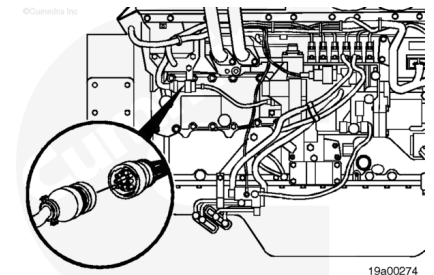
19a00558

Connect the engine harness to the left bank fuel shutoff valve connectors.



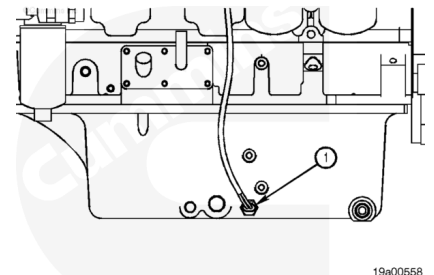
19a00249

Connect the engine harness to the left bank fuel pump connector.



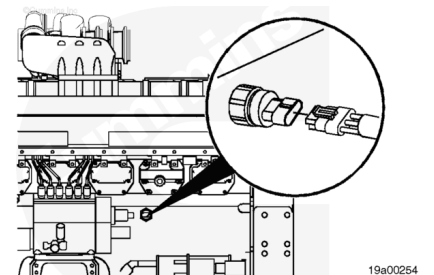
19a00274

Connect the engine harness to the oil temperature sensor, if equipped.



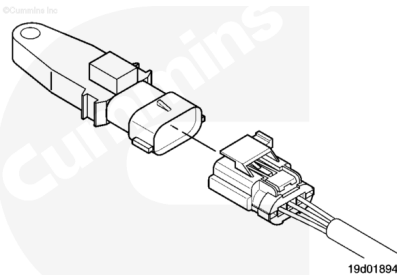
19a00558

Connect the engine harness to the oil pressure sensor.

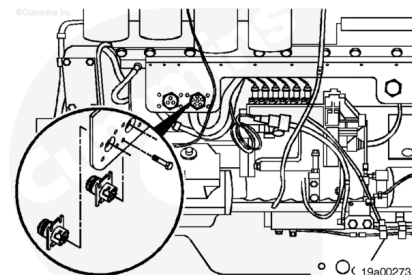


19a00254

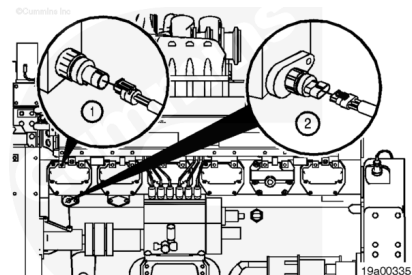
Connect the inline connector bracket which includes the barometric pressure sensor connector.



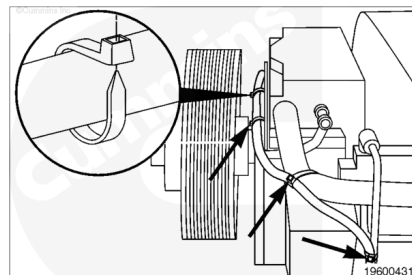
Connect the two 6-pin or one 9-pin data link connector(s) to the harness support bracket by installing the capscrews.



Connect the engine harness to the left bank intake manifold pressure sensor.



Install the ties on the thermostat housing bracket, aftercooler tube, and fuel tube.

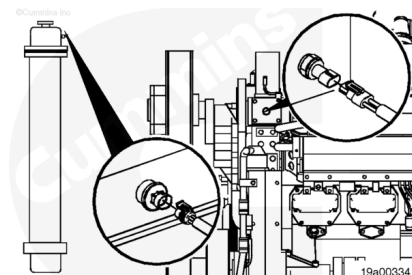


Connect the engine harness to the coolant level harness extension connector.

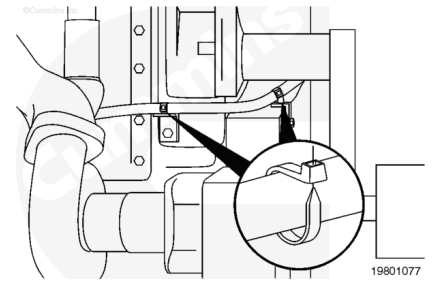
Connect the engine harness to the coolant temperature sensor and the coolant level sensor.

Connect the engine harness to the coolant pressure sensor, if equipped.

Install the engine harness clamps and ties.

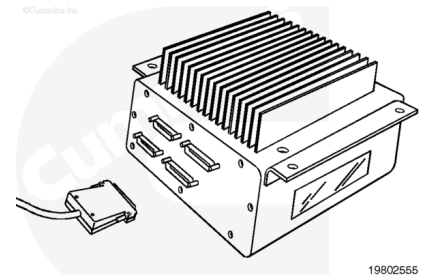


Connect the engine harness to the engine using cable ties.



Connect the engine harness Deutsch connector to the engine harness extension cable, if used, or the engine harness adapter cable. Use a 4 mm [5/32 in] hex head wrench to tighten.

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.

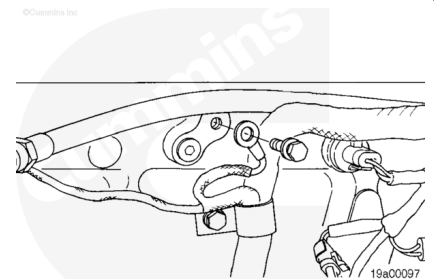


Torque Value: 2.8 n•m [25 in-lb]

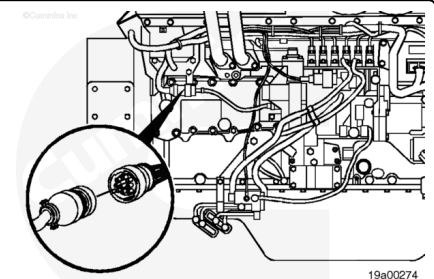
Note : Do not over-torque connector as damage can occur.

Connect the engine block ground to the engine block.

Torque Value: 20 n•m [177 in-lb]

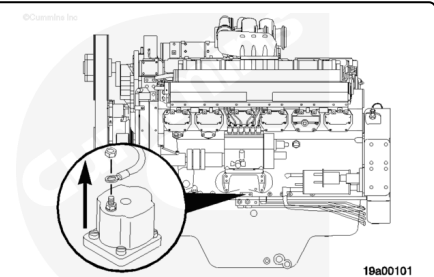


Connect the engine harness to the fuel pump connectors (left bank and right bank).

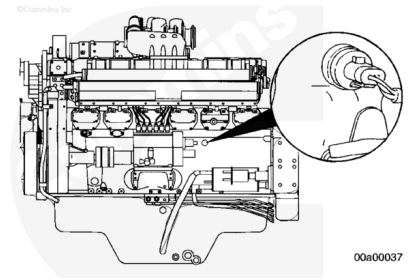


Connect the fuel shutoff control wires to the fuel shutoff solenoids (left bank and right bank). Tighten the retaining nut.

Torque Value: 3 n•m [27 in-lb]

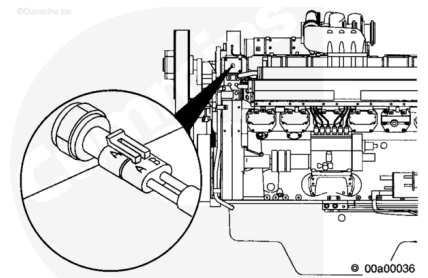


Connect the engine harness to the oil pressure sensor.

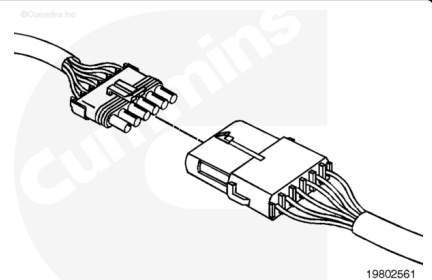


Connect the engine harness to the coolant temperature sensor.

Connect the engine harness clamps.



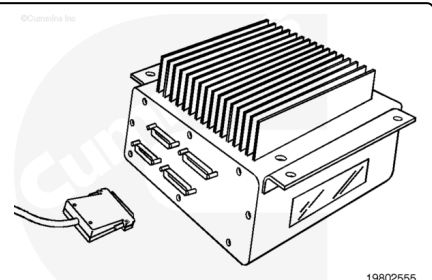
Connect the engine harness to the electrical system at the Inline E connector.



Connect the engine harness extension cable to the extension cable connectors.

Connect the engine harness adapter cable and all engine harness extension cables used.

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.



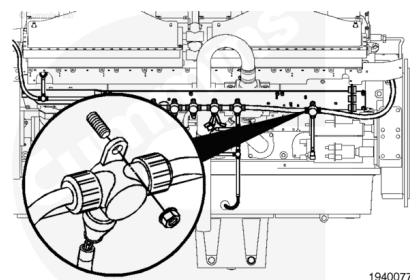
Torque Value: 2.8 n·m [25 in-lb]

Note : Do not over-torque connector as damage can occur.

QSK45, QSK60, and QSK78

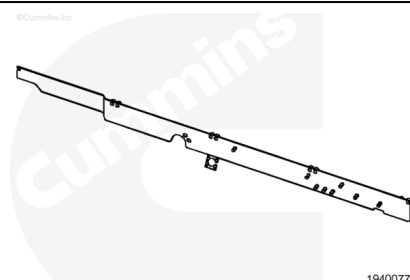
Note : This step can not be completed on the QSK78 engine. The engine harness and the engine support bracket are one piece.

Connect the engine harness to the engine harness support bracket.

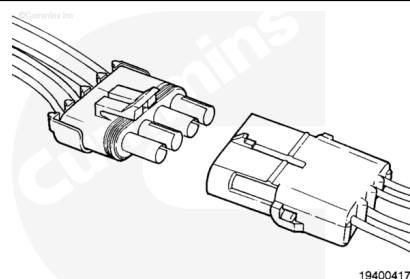


Install the engine harness and engine harness support bracket onto the engine.

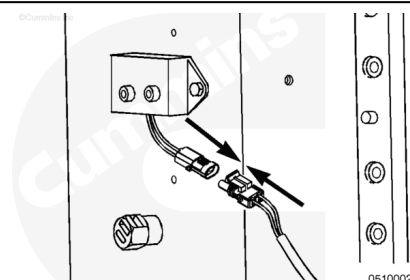
Torque Value: 45 n•m [33 ft-lb]



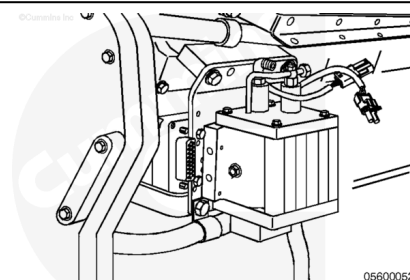
Connect the coolant level switch sensor at the 4-pin inline connector.



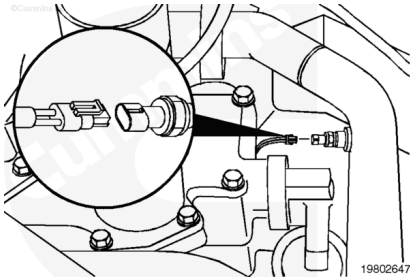
Connect the engine harness to the Centinel™ makeup tank oil level sensor Inline connector, if equipped.



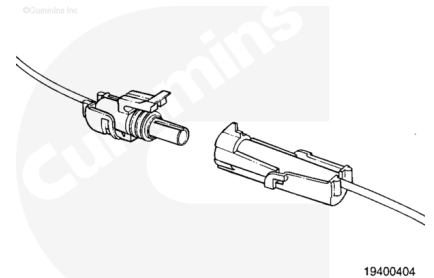
Connect the engine harness to the Centinel™ actuator connector, if equipped.



Connect the engine harness to the aftercooler water temperature sensor.



Connect the engine harness to the flash status at the Inline H connector.



Note : There are a number of different versions of the breather box QSK45, QSK60, and QSK78 engines. Even though the appearance can be different, the same steps can be used to install all versions of the breather box.

Install the breather box assembly (8).

Install the six mounting capscrews into the breather box assembly.

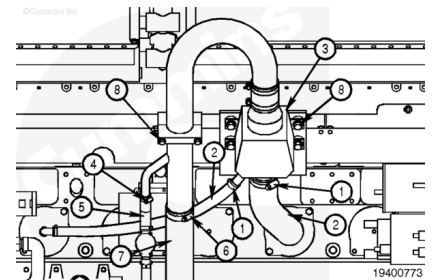
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

Connect the lower breather hose (7) to the breather box using the hose clamp (6).

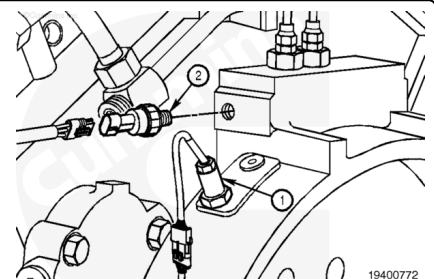
Connect the blow by pressure sensor/hose assembly (5) to the breather box using hose clamp (4).

Connect the three hoses (2) to the bottom of the breather box (3), using three hose clamps (1).



Connect the engine harness to the engine speed sensor (1).

Connect the engine harness to the oil pressure sensor (2).

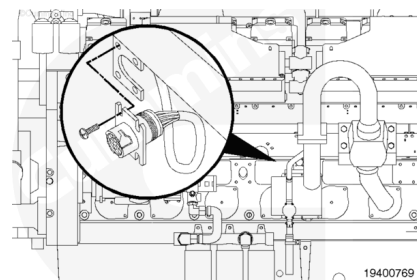


Note : This step can **not** be completed on the QSK78 engine. The engine harness and the engine support bracket are one piece.

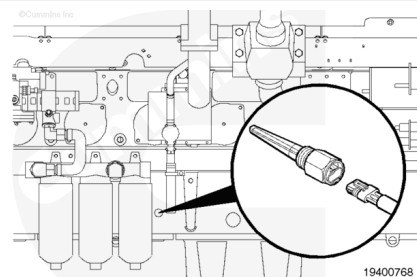
Slide the 9-pin data link connector into the mounting slot in the engine harness support bracket.

Install and tighten the capscrews.

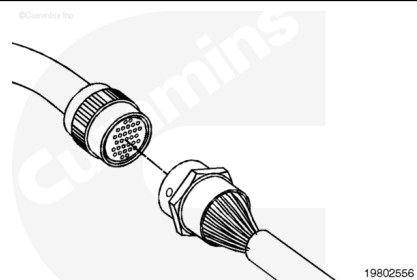
Torque Value: 1 n•m [9 in]



Connect the engine harness to the oil level sensor, if equipped.



Connect the engine harness to the extension harness at the Inline A, Inline B, and Inline C connectors.

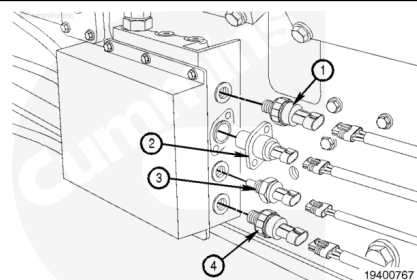


Connect the engine harness to the timing rail pressure sensor (1).

Connect the engine harness to the fuel temperature sensor (3).

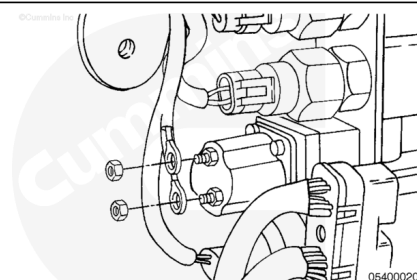
Connect the engine harness to the injector meter rail pressure sensor (4).

Connect the engine harness to the barometric pressure sensor (2). This sensor can **not** be located on the electronic control valve assembly. Locate the sensor on the harness main trunk/support bracket.



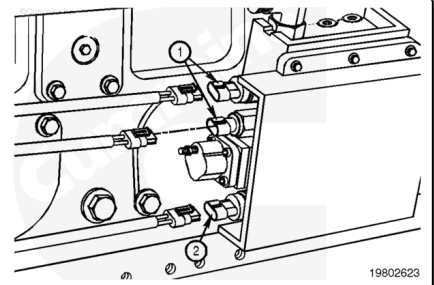
Connect the engine harness to the fuel shutoff valve.

Torque Value: 3 n•m [27 in-lb]



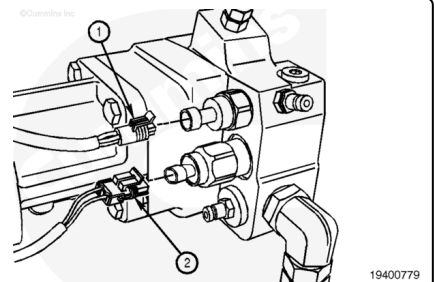
Connect the engine harness to the fuel system actuators.

1. Timing actuators
2. Fueling actuator.

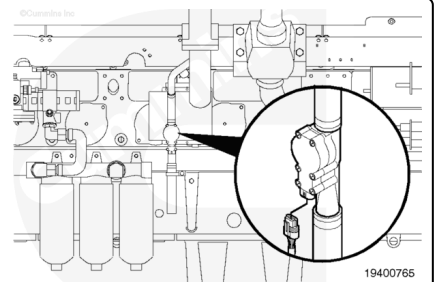


Connect the engine harness to the fuel pump pressure sensor (1).

Connect the engine harness to the fuel pump actuator (2).

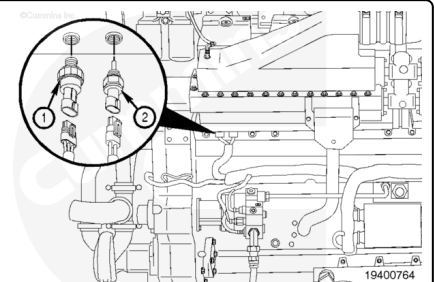


Connect the engine harness to the blowby sensor.



Connect the engine harness to the intake manifold pressure sensor (1).

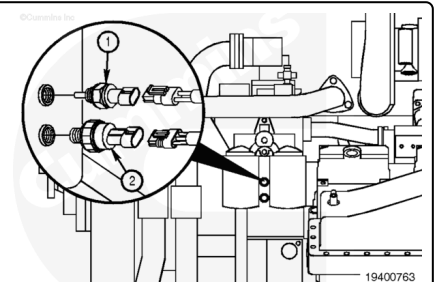
Connect the engine harness to the intake manifold temperature sensor (2).



The coolant pressure sensor can be in three different locations:

- Between the coolant filters, as illustrated.
- In the block, behind the fuel pump.
- In a spacer block between the coolant filters, and the thermostat housing.

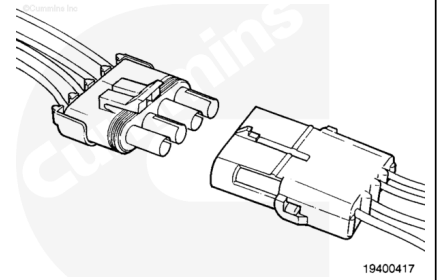
Connect the engine harness to the coolant pressure sensor.



Connect the engine harness to the Inline E connector.

Make sure all engine harness connections are secure.

Connect the electronic service tool and check for fault codes.



Last Modified: 23-Apr-2014
